ENGINEERING SCIENCE

Revised Feb 2022

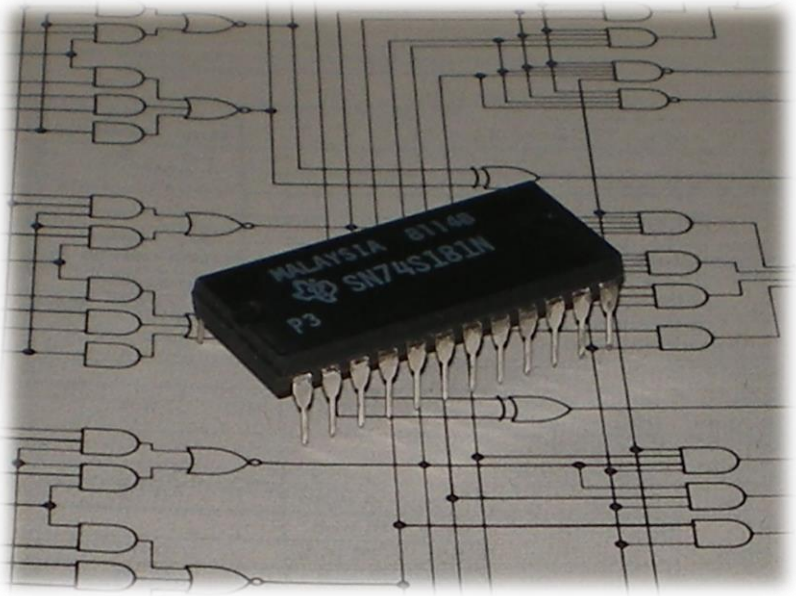
Pupil booklet

Digital logic

Topic 1: Truth tables

Topic 2: Boolean

Topic 3: Combinational logic



Name: _____

Class: _____

Teacher: _____

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Learning Intentions

- To learn about "Logical" electronic systems
- To know what an IC chip is and how it is used
- To know about simple and complex logic gates
- To know and understand what a truth table is
- To know what a Boolean expression is and how to develop it

Success Criteria

Use traffic lighting to record your progress. This will also help with revision as you will be able to see what areas you need to work on. Your teacher will sign off this page after speaking to you about your self-evaluation:

Red: I need to work on this topic further to understand it and ask for help.
Amber: I need a bit of help to fully understand this topic.
Green: I am confident in my understanding of this topic.

- ☐ I can draw circuit diagrams of logic circuits.
- ☐ I can develop analogue electronic control systems.
- ☐ I can develop digital electronic control systems.
- ☐ I am able to Simulate and construct digital control systems using logic components.
- ☐ I understand the terms logic 1 and logic 0.
- ☐ I can draw truth tables for the following gates: AND, OR, NOT.
- ☐ Using simple formulae, I can construct Boolean equations.

Why?

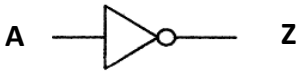
Digital electronics play an increasingly essential role in the way we lead our lives. Computer operations rely on logic (these are included in your phone, television and other household appliances). Likewise, machine operation and simple design of electronic circuits are design using logic. This means that we can begin "making decisions" through engineered systems.

Switching logic

Although it may not always seem like it, electronics and electronic systems are very logical in the way that they work. In the simplest form, if you want a light to come on, then you press a switch. Of course, it gets more complicated than that. Most technological systems involve making more complicated decisions: for example, sorting out bottles into different sizes, deciding whether a room has a burglar in it or not, or knowing when to turn a central heating system on or off.

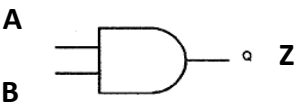
Logic gates are used in dealing with and processing a combination of different inputs. This switching logic can be applied to electrical switches and sensors, pneumatic valves or hydraulic systems. Switching logic uses logic gates to perform decisions.

NOT



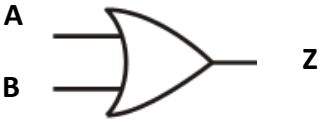
When a high input signal rails (A), a low output (Z) is recorded.
When a low input signal rail (A), a high output signal is recorded.

AND



Only when both input signal rails (A,B) are high does the output rail (Z) become high. Otherwise, output remains low (off / O).

OR



When either or both input signal rails (A,B) are high, the output rail (Z) is low.

The easiest way to represent how each gate behaves is to make use of truth tables. A truth table shows all possible combinations of inputs and outputs to a logic gate.

*Digital signals can be at a HIGH voltage level or a LOW voltage level. In logic circuits a LOW signal is said to be at logic '0' or **off** and a HIGH signal at logic ' 1' or **on**.*

The number of combinations in one column will conform to this rule:

2 is the number of inputs.
Hence, $2^2 = 4$. So Input A has 4 lines. Half of which are 0, the other half are at logic state 1.

Input A	Input B
0	0
0	1
1	0
1	1

Truth tables

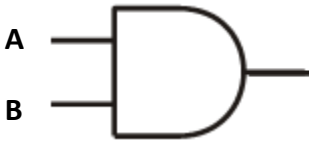
An example of gate logic truth tables are shown below. Only NOT, AND, OR gates are used at National 5. The other gates (NAND, NOR, EOR) are investigated further at Higher Engineering Science but it is good to be introduced to them earlier.

NOT GATE



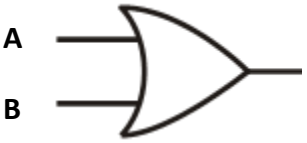
A	Z
0	1
1	0

AND GATE



A	B	Z
0	0	0
0	1	0
1	0	0
1	1	1

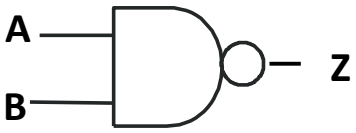
OR GATE



A	B	Z
0	0	0
0	1	1
1	0	1
1	1	1

NAND Logic Gate

The NAND gate is effectively an *inverted* AND gate. In other words, the results obtained from the output are the opposite to those of the AND gate. This gate is often referred to as 'NOT AND'.

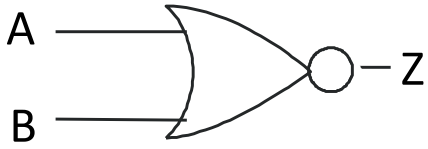


When drawing up the truth table for the NAND gate it can be difficult to 'picture' or imagine the results. The best way to do this is to pretend that it is an AND gate and then invert (reverse) the results, thus giving you the outputs for the NAND gate.

1. Copy and complete the symbol and truth table into your jotter.

A	B	*AND*	Z (NAND)
0	**0**		
0	**1**		
1	**0**		
1	**1**		

NOR Logic Gate



The NOR gate is effectively an inverted OR gate. In other words, the results obtained from the output are the opposite to that of the OR gate. This gate is often referred to as 'NOT OR'.

As with the NAND gate, when drawing up the truth table for the NOR gate it can be difficult to 'picture' or imagine the results. The best way to do this is to pretend that it is an OR gate and then invert (reverse) the results, thus giving you the outputs for the NOR gate.

2. Copy and complete the symbol and truth table into your jotter.

A	B	*OR*	Z (NOR)
0	**0**		
0	**1**		
1	**0**		
1	**1**		

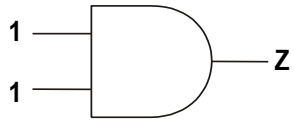
XOR Logic Gate

This IC sends out logic when only 1 OR the other is sending logic 1. It will not work if both signals are sending out logic 1 like a normal OR gate does. This is why a XOR gate is sometimes known as an 'X-treme OR' or an 'Exclusive OR'

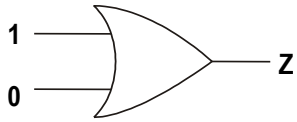


3. Copy each logic diagram into your jotter and state whether the output Z is at logic 0 or logic 1. It would be useful to create a truth table for each gate system to calculate your answer. Prove your answer on Yenka.

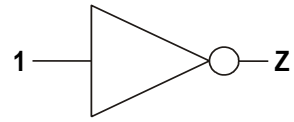
(a)



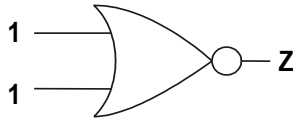
(b)



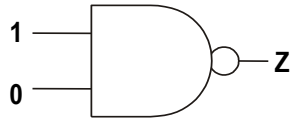
(c)



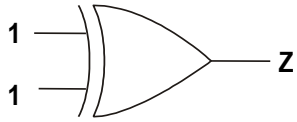
(d)



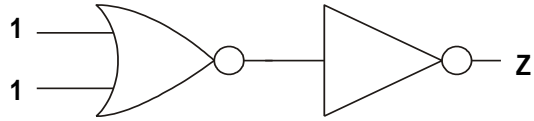
(e)



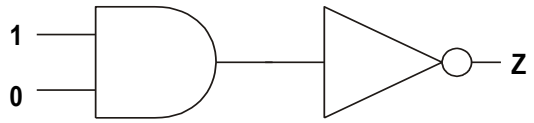
(f)



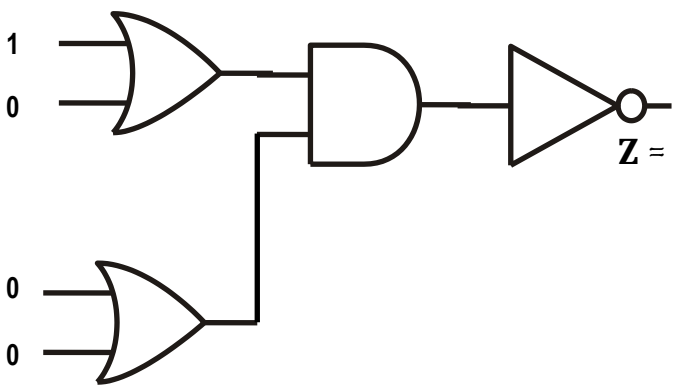
(g)



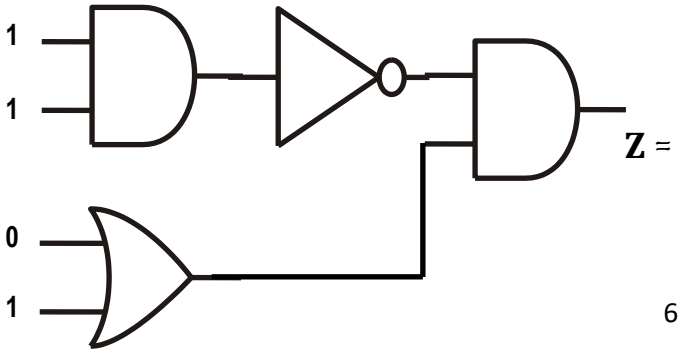
(h)



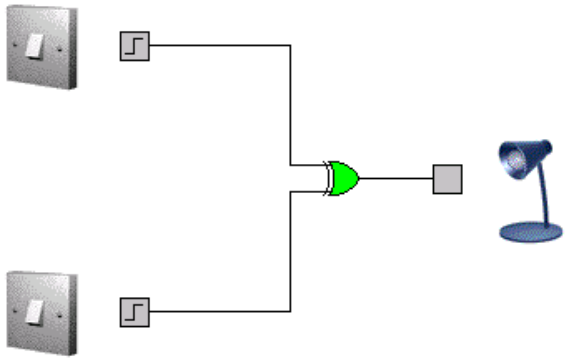
(i)



(j)

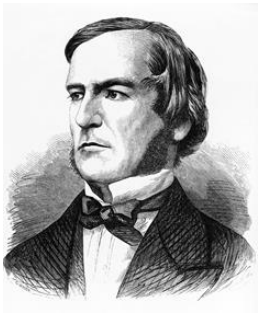


4. Use Yenka to set up the system shown below. Copy and complete the symbol and truth table into your jotter. In Yenka, follow the folder pathway of: 'PCB's with electronics', 'Electronic components', 'Digital processing' then 'Digital 7400 series'. Animated inputs (switch) and outputs (lamp) are located in the folder under 'PIC's'. Use the 7486 exclusive – Or gate. The circuit does not require a power supply.



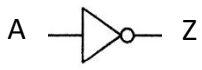
A	**B**	OR	**Z** (XOR)
0	**0**		
0	**1**		
1	**0**		
1	**1**		

Boolean Expressions

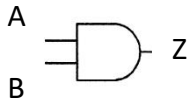


A man called George Boole (1815-1864) was a largely self-taught mathematician. He wanted a mathematical way to represent logic systems. This is the man to blame for what you are about to learn! He devised Boolean Algebra which is still used to this day to assist the designing of logic systems.

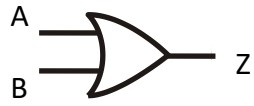
Each logic gate has a corresponding Boolean mathematical formula or expression. The use of these expressions saves us having to draw symbol diagrams over and over again.

NOT 

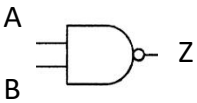
$Z = \overline{A}$

AND 

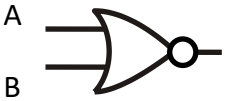
$Z = A.B$

OR 

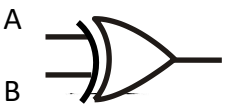
$Z = A+B$

NAND 

$Z = \overline{A.B}$

NOR 

$Z = \overline{A+B}$

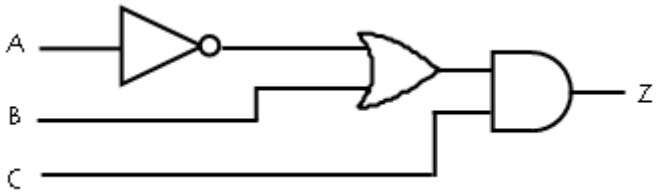
XOR 

$Z = A \oplus B$

Combinational Boolean

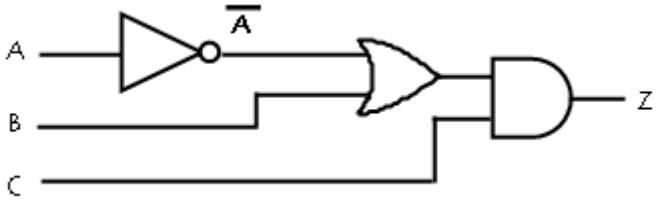
So far, we have only looked at simple logic systems. In reality, most logic systems use a combination of different types of logic gates. This is known as 'Combinational Logic'. Boolean Expressions can be worked out from these to know the equation for the circuit.

Example



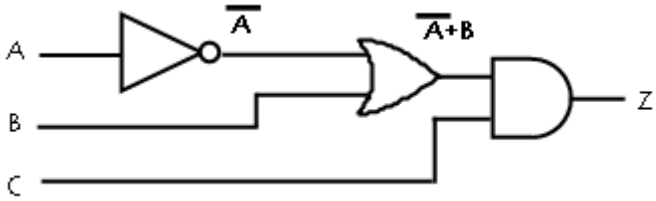
We must work this out one step at a time, creating an equation as the signal passes through each logic gate:

STEP 1:



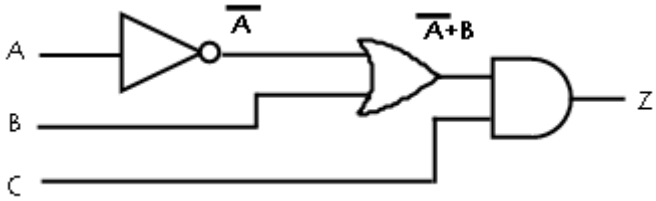
You can find out that the line for A soon turns into $\overline{A}$

STEP 2:



The lines for A and B now changes at it goes through the next logic gate.

STEP 3:



As we progress through the circuit we can now add C.

The result is $Z = (\overline{A}+B).C$

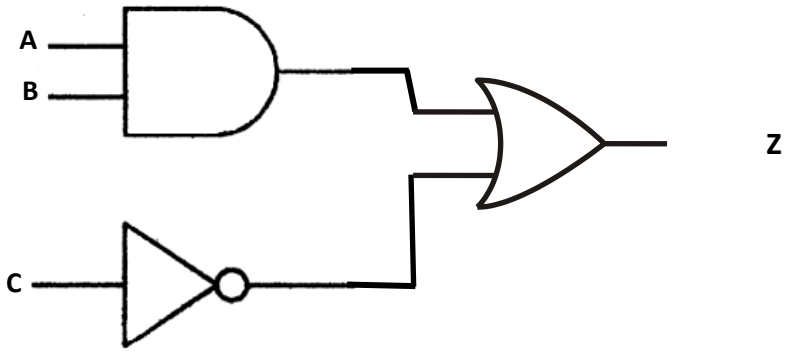
This 'Dot' means Z only operates after the inclusion of the 'AND' gate.
Z = Inverted A 'OR' B 'AND' C

5. Copy each diagram into your jotter and work out the Boolean Expression for the following:

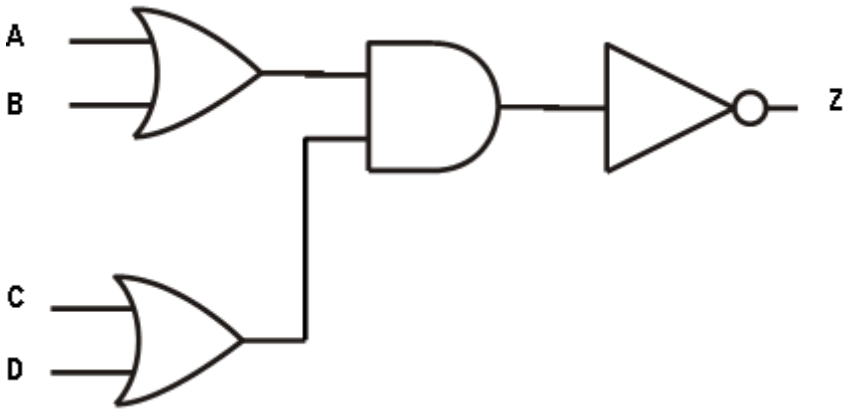
A)



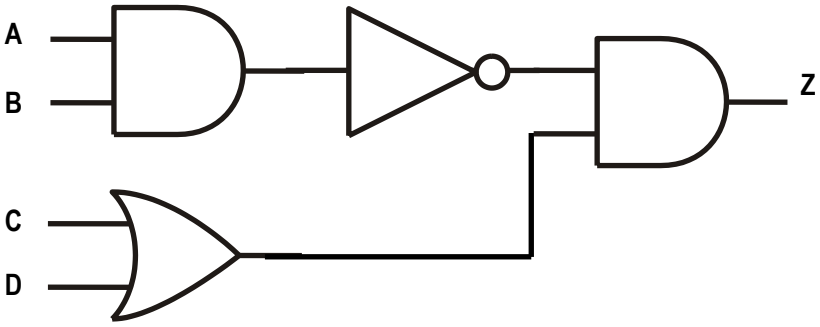
B)



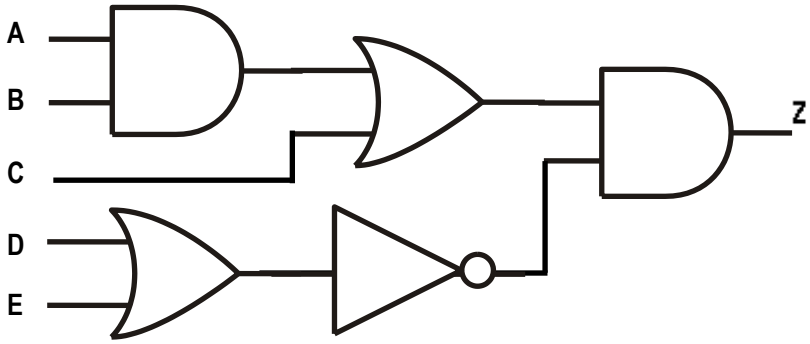
C)



D)



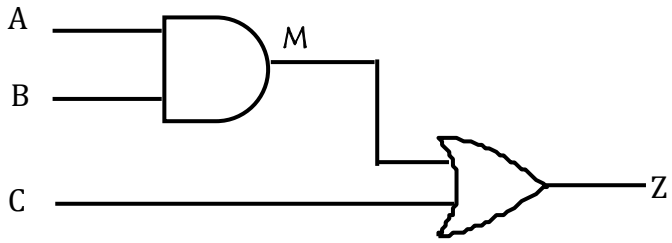
E)



Combinational logic tables

For each combinational logic diagram a logic table can also be worked out. This can look confusing but if we take our time, it can be very simple.

Example



By creating a new row in our truth table for once the circuit has went through a logic gate we can take this in simple steps

Step 1:

We also need to have extra rows to ensure we are showing **EVERY** possibility.

In this case we need to work out 2 possible options for everything the OR gate. This means there will be 8 possible options.

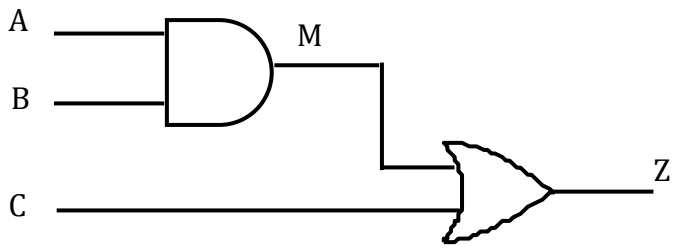
Another we of ensuring we have all the possible options is using the Say we have 3 inputs to a logic system, using powers of 2 we can calculate the number of possible combinations of input to the circuit as follows,

No. of combinations = 2^3 = 8 rows as shown opposite

A	B	C	M	Z

Example: 4 inputs would mean '2' to the power of '4': 2^4 = 16 rows

Step 2:



We now have to take this in stages.
M is the output for A and B so this is what to work out first.

A	B	C	M	Z
0	0		0	
0	0		0	
0	1		0	
0	1		0	
1	0		0	
1	0		0	
1	1		1	
1	1		1	

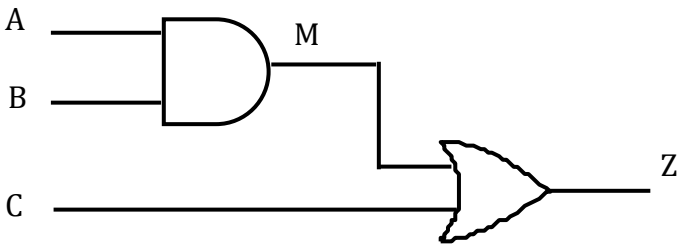
Step 3:

We now have to put in the options for C ensuring we have covered ever single possibility.

Eg. There should be two 0 – 0 combinations for A&B, so we have to put the 2 possible options in for C to ensure we have covered every single possible option

A	B	C	M	Z
0	0	0	0	
0	0	1	0	
0	1	0	0	
0	1	1	0	
1	0	0	0	
1	0	1	0	
1	1	0	1	
1	1	1	1	

Step 4:

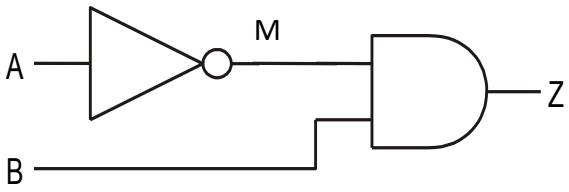


By looking at our diagram, we can see the only inputs to Z is D & C.
We now use these 2 rows to work out Z

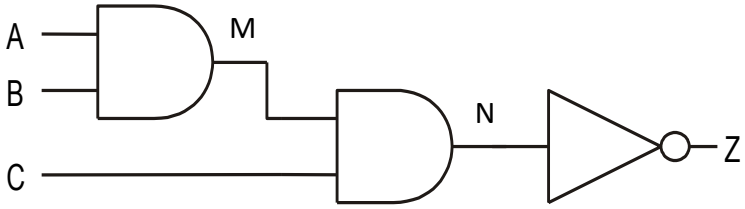
A	B	C	M	Z
0	0	0	0	0
0	0	1	0	1
0	1	0	0	0
0	1	1	0	1
1	0	0	0	0
1	0	1	0	1
1	1	0	1	1
1	1	1	1	1

6. In your jotters, work out the truth tables and Boolean expressions for these circuits.

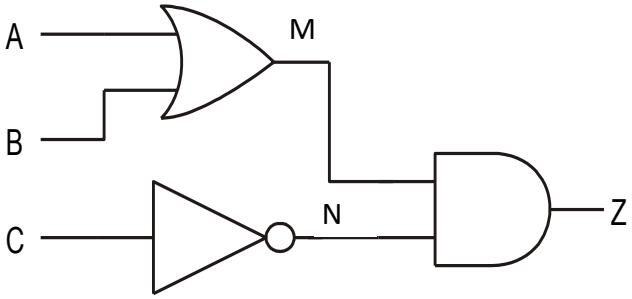
A)



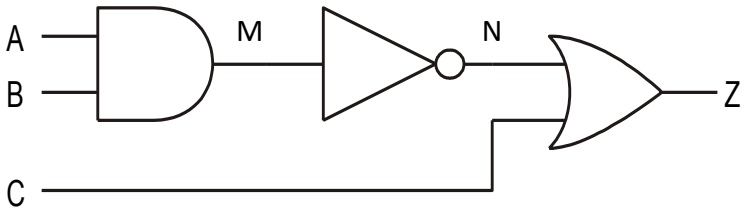
B)



C)



D)



Creating Logic Diagrams from Truth Tables

When designing systems, it is normal to design a logic diagram from a prepared truth table. This may seem difficult to start with, but if you concentrate on the combinations which give a logic 1 condition in the output column, solutions can be found easily.

The truth table below shows two inputs, A and B, and one output, Z.

A	B	Z
0	0	0
0	1	0
1	0	1
1	1	0

$$Z = A \cdot \overline{B}$$

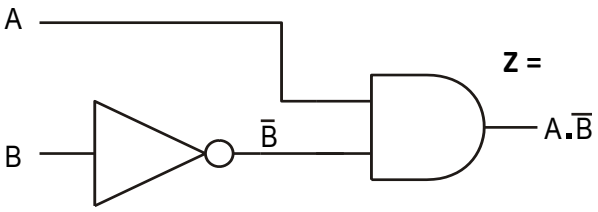
The output Z is at logic 1 in the third row down, and we can see that for this to happen A must be at logic 1 and B must be at logic 0. In other words:

$$Z = A \textbf{ AND } \text{NOT } B$$

This means that we need a two-input AND gate, with B being fed through a NOT gate. We can write the statement in shorthand Boolean as:

$$Z = A.\overline{B}$$

Therefore:



Example

In this problem below, we have three inputs, A, B and C, with one output, Z. From the truth table we can see that there are two occasions when the output goes to logic 1.

A	B	C	Z
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

$$Z = \overline{A}.B.C$$

$$Z = A.B.\overline{C}$$

$$Z = (\overline{A}.B.C) + (A.B.\overline{C})$$

The answer continues on the next page.

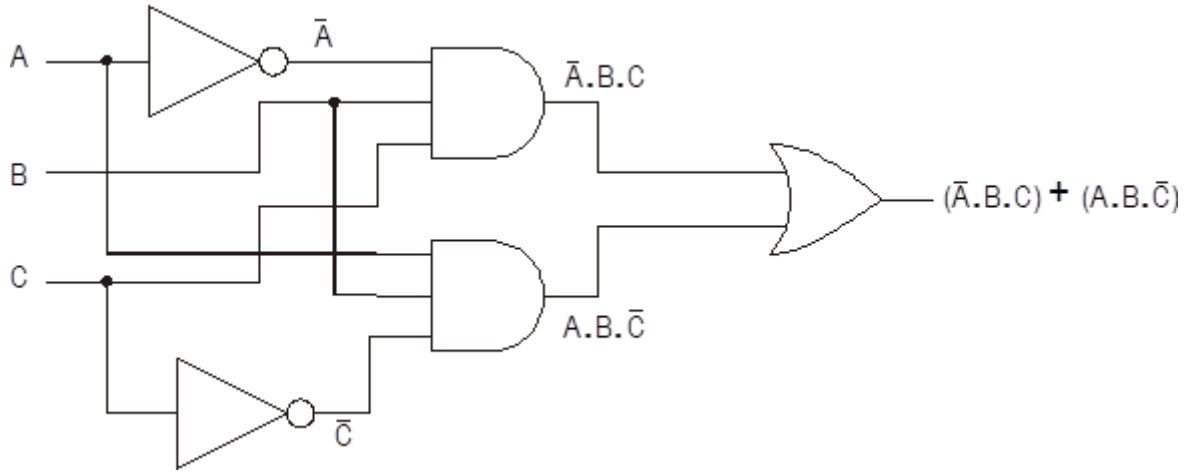
In other words, Z = 1 if

A is at logic 1 AND B is at logic 1 AND C is at logic 1

OR

A is at logic 1 AND B is at logic 1 AND C is at logic 0

This means we need a two-input OR gate being fed from two three-input AND gates as shown below.

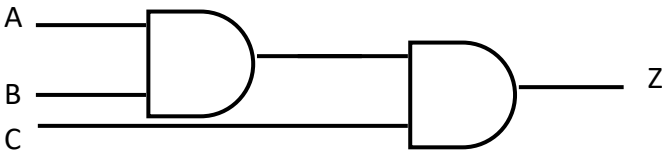


The shorthand Boolean equation for this truth table is

$$Z = (A.B.C) + (A.B.C)$$

Triple AND gates

The example above was created by using an AND gate with 3 input rails. This can also be illustrated using two AND gates to provide you the same answer.



7. In your jotters, work out the Boolean expression and draw the logic diagram for each of these truth tables. Include the truth table and use a ruler for all your work.

A	B	Z
0	0	0
0	1	1
1	0	0
1	1	0

(a)

A	B	Z
0	0	1
0	1	0
1	0	1
1	1	0

(b)

A	B	Z
0	0	0
0	1	1
1	0	1
1	1	0

(c)

8. In your jotters, work out the Boolean expression and draw the logic diagram for each of these truth tables: Include the truth table and use a ruler for all your work.

A	B	C	Z
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

(d)

A	B	C	Z
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	0

(e)

A	B	C	Z
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

(f)

Integrated circuits

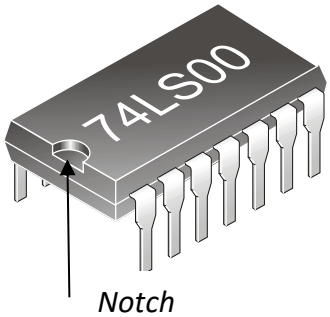
Although logic gates have electronic symbols, they are not discrete components: they are contained in integrated circuits. Integrated circuits (ICs) are silicon-based components containing complex circuits. There are two main types TTL (Transistor-Transistor Logic or 7400IC series) or CMOS (Complementary Metal Oxide Silicon or the 4000IC series). You will be dealing with TTL IC chips and their reference numbers start with "74". TTL stands for Transistor – Transistor logic.

In Yenka, follow the folder pathway of: 'PCB's with electronics', 'Electronic components', 'Digital processing' then 'Digital 7400 series'. Animated inputs and outputs are located in the folder under 'PIC's'.

You can see the outline diagram of the 74 serious in your data booklet. This diagram will show you the pin connections as the pins. IC's are too small to have the numbers on in real life.

Key things to remember are that there will be a notch in one end of the IC and beside it a big dot. The dot is positioned next to Pin 1 and the pins are numbered anti-clockwise from pin1 to pin14.

Notice that the IC gets its power through Pin 7 and Pin 14 and it is a signal voltage going to or coming from the other pins.



Differences between TTL and CMOS ICs

TTL chips have been traditionally used in schools because CMOS chips were easily damaged by static energy. However, there are other differences.

CMOS chips only use small currents (about 8uA) while TTL ones use a much larger current (about 3mA, which is almost 400 times as much). This is reflected in their power consumption especially when you may have hundreds of them in a mainframe computer.

TTL chips need more precise supply of voltage than their CMOS equivalents but they have much faster switching speeds, which is important in high-speed telecommunications, etc.

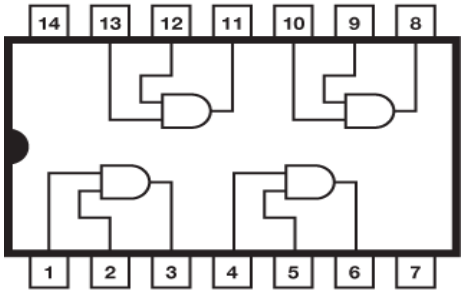
The main differences are summarised in the table below.

Property	**TTL**	**CMOS**
Series	7400	4000
Power supply	5 + or – 0.25v	3 to 15v
Unused inputs	Float high	indeterminate
Power consumption	High	Low
Switching speed	Fast	Slow
Effect of static	Not affected	Can be damaged

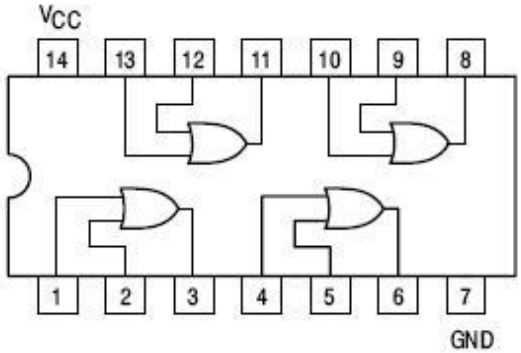
Building Circuits with IC Chips

Although the chip contains complex circuitry, the internal wiring can be shown as simple logic circuits with the inputs and outputs of each logic gate shown. This is called a *pin-out diagram*. The IC type is shown underneath the IC image.

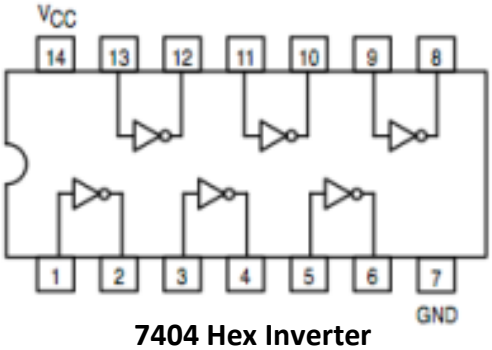
Pin 14 is connected to the 5-volt stable supply (power supply) and pin 7 to 0 volts (ground line). This must ALWAYS be done to ensure the chip is powered.



7408 Quad2 input AND Gate



7432 Quad2 input OR Gate



7404 Hex Inverter

Chip name: Number of Gates (4 or Quad) then number of inputs (2) and then type of Gate.

Test Equipment

When trying to establish logic levels within a complex system or to monitor a logic output without using an LED, we use a digital logic probe.



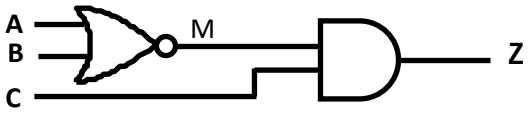
The logic probe is powered from the same supply as the logic circuit being tested and the needle point is pushed against the various pins on the IC to test their logic level. Normally the logic probe gives out a high-pitched sound and a red LED lights if the pin being tested is at logic 1. If the point tested is at logic 0, a low-pitched sound is emitted and a green LED lights.

9. Work out the logic for these circuits. Use Yenka to test your answers. Use the tables over the page to help you complete the tables.

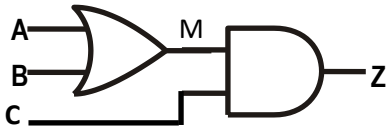
a)



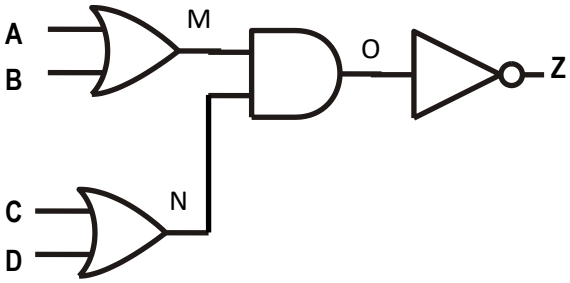
b)



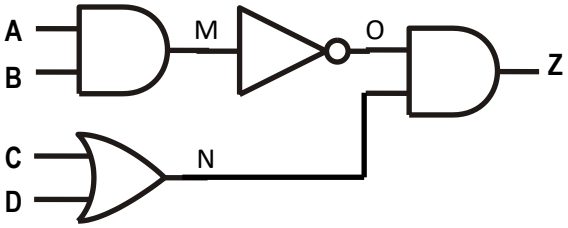
c)



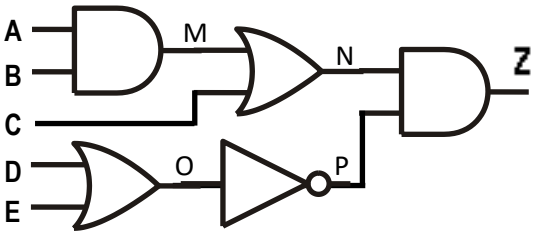
d)



e)



f)



a)

A	B	M	Z
0	0		
0	1		
1	0		
1	1		

b)

A	B	M	C	Z
0	0		0	
0	1		0	
1	0		0	
1	1		0	
0	0		1	
0	1		1	
1	0		1	
1	1		1	

c)

A	B	M	C	Z
0	0		0	
0	1		0	
1	0		0	
1	1		0	
0	0		1	
0	1		1	
1	0		1	
1	1		1	

d)

A	B	M	C	D	N	O	Z
0	0		0	0			
0	1		1	1			
0	0		1	1			
0	1		0	0			
0	0		0	0			
0	1		1	1			
0	0		1	1			
0	1		0	0			
1	0		0	1			
1	1		1	0			
1	0		1	0			
1	1		0	1			
1	0		0	1			
1	1		1	0			
1	0		1	0			
1	1		0	1			

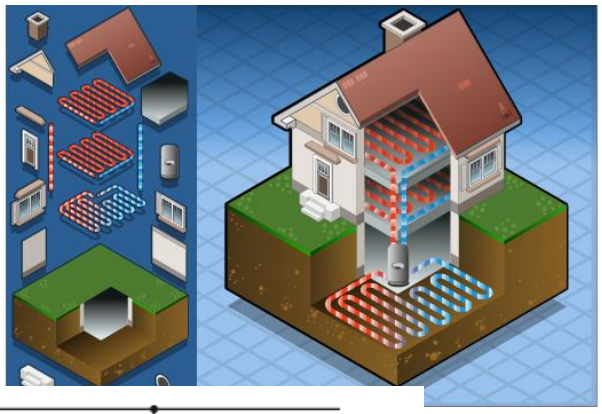
e)

A	B	M	C	D	N	O	Z
0	0		0	0			
0	1		0	1			
0	0		1	0			
0	1		1	1			
0	0		0	1			
0	1		0	0			
0	0		1	1			
0	1		1	0			
1	0		1	0			
1	1		1	1			
1	0		0	1			
1	1		0	0			
1	0		1	1			
1	1		1	0			
1	0		0	0			
1	1		0	1			

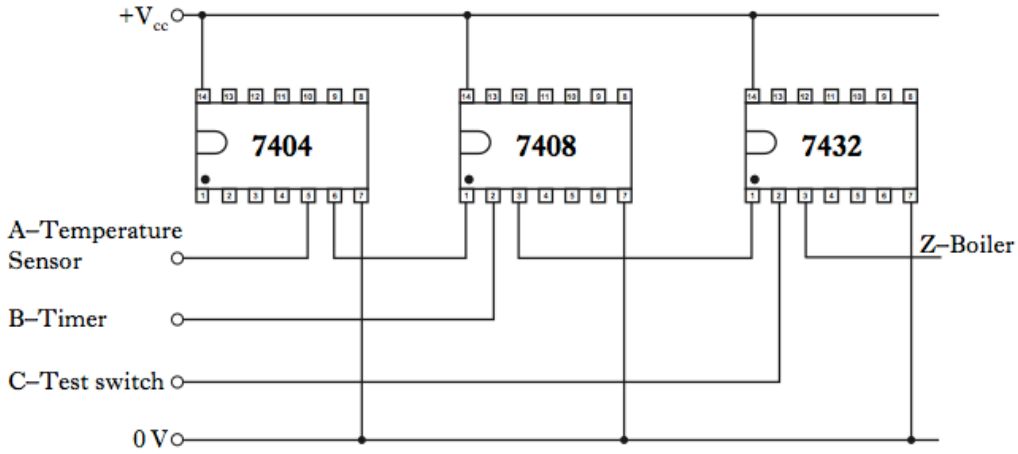
f)

A	B	M	C	N	D	E	O	P	Z
0	0		0		0	0			
0	1		0		1	0			
0	0		1		1	1			
0	1		1		0	1			
0	0		0		1	1			
0	1		0		0	1			
0	0		1		0	0			
0	1		1		1	0			
0	0		0		0	0			
0	1		0		1	0			
0	0		1		1	1			
0	1		1		0	1			
0	0		0		1	1			
0	1		0		0	1			
0	0		1		0	0			
0	1		1		1	0			
1	0		0		0	0			
1	1		0		1	0			
1	0		1		1	1			
1	1		1		0	1			
1	0		1		1	1			
1	1		0		0	1			
1	0		1		0	0			
1	1		1		1	0			
1	0		0		0	0			
1	1		0		1	0			
1	0		1		1	1			
1	1		1		0	1			
1	0		0		1	1			
1	1		0		0	1			
1	0		1		0	0			
1	1		1		1	0			

10. The central heating in a house is to be controlled by an electronic circuit.



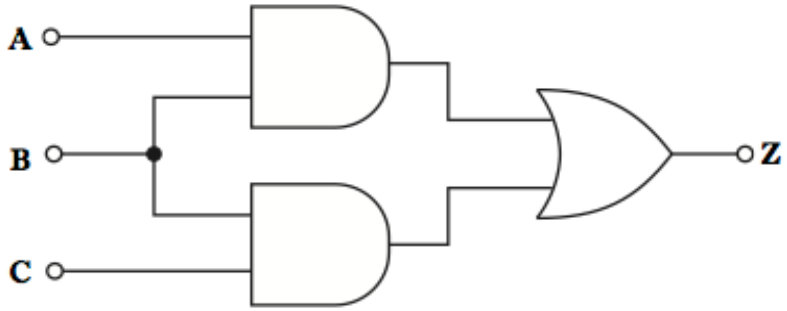
Part of the electronic circuit is shown below.



Complete, with reference to the 7400 pin-out diagram and the wiring diagram, the logic diagram for the central heating system. Use page 17 of this booklet for 'IC' reference.



11. The following logic diagram is required for an electronic alarm system.



a) Develop the Boolean expression for Z, in terms of A, B and C.

Z =

b) Complete the truth table below for the logic diagram.

A	B	C	Z
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	
1	0	0	0
1	0	1	
1	1	0	
1	1	1	1

12. The Boolean statement used to describe part of the operation of a coffee machine is shown.

$$(\bar{A} \cdot B) + (A \cdot B) = Z$$

(*a*) Complete the truth table below, for the Boolean statement.

A	B	Z
0	0	
0	1	
1	0	
1	1	

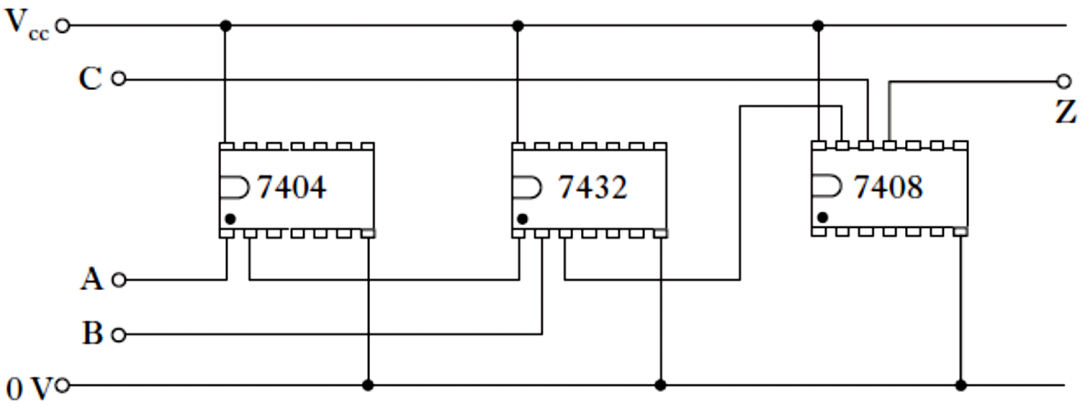
(*b*) Complete, with reference to the Boolean statement, the logic diagram for the coffee machine using NOT, AND and OR gates.



The circuit is to be constructed from the 4000 series family of Integrated Circuits (ICs).

(*c*) State the name of this IC logic family.

13. The wiring diagram for a digital electronic circuit is shown



(*a*) Complete, with reference to the above diagram and the Data Booklet, the logic diagram for the circuit.



b) State, with reference to the data booklet, the **full** name of the following Integrated Circuits.

7432 -

7408 -

The circuit uses the TTL logic family.

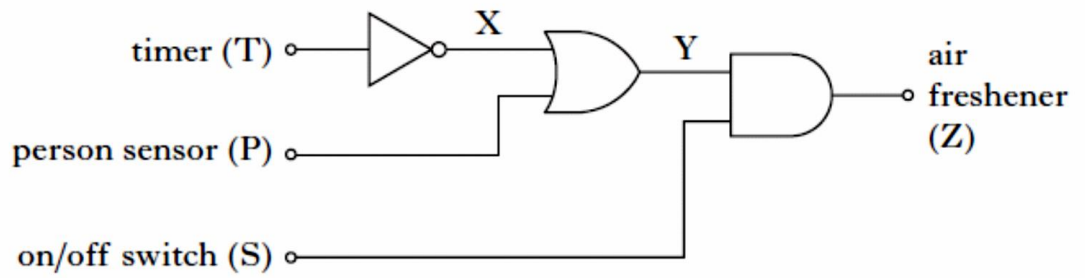
(*c*) State, for this logic family:

(i) its full name;

(ii) its operating voltage;

(iii) one reason, other than operating voltage, for its use.

14. The logic diagram for an electric air freshener is shown



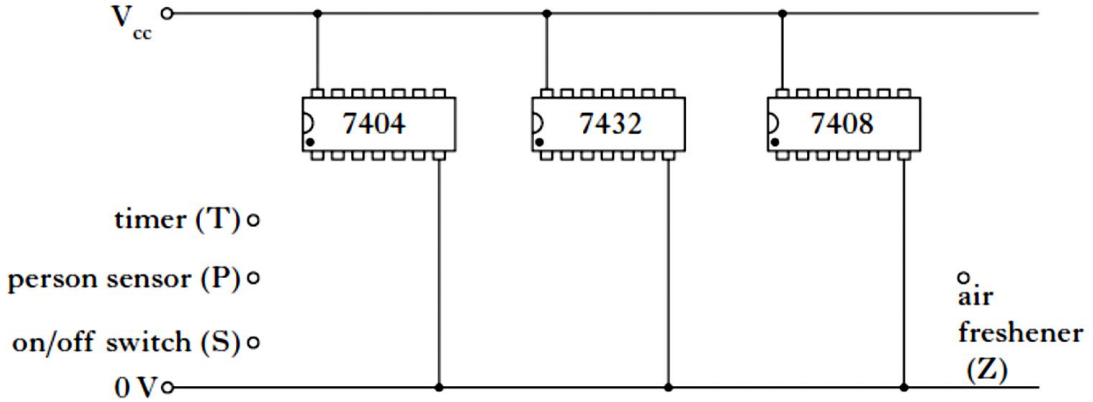
a) State the Boolean expression for the air freshener (Z) in terms of T, P and S.

Z =

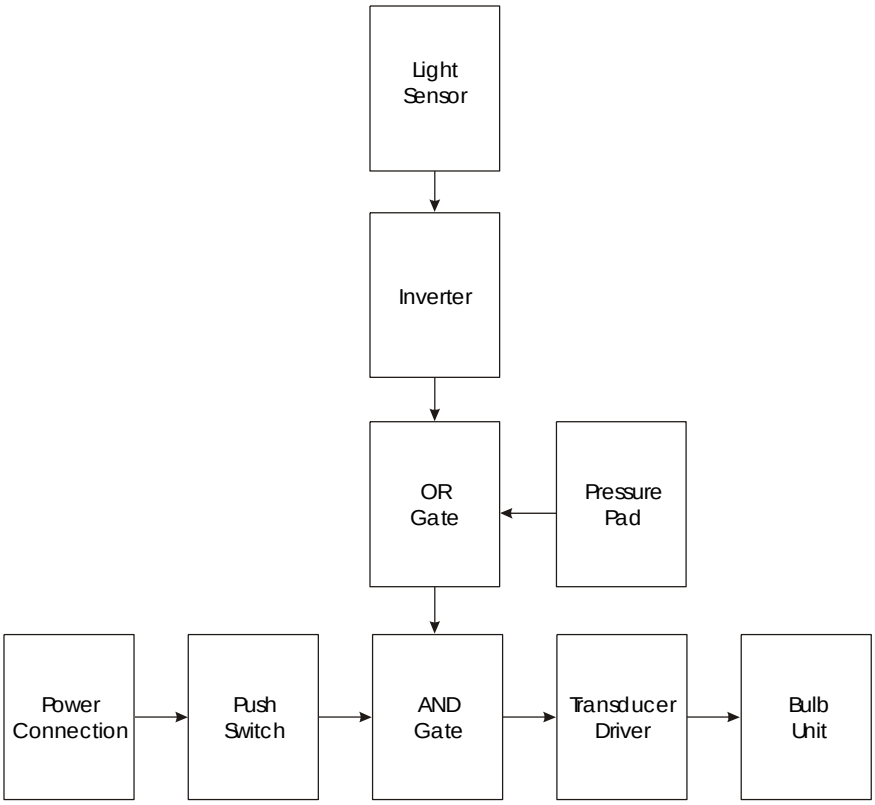
b) Complete the truth table for the logic diagram shown

S	P (1 = person sensed)	T (0 = timer on)	X	Y	Z
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

c) Complete the wiring diagram below.



15. a) Below is a circuit, shown using modular boards. They are easy to understand, as they have the names of the parts of the circuit on them.



b) Draw a logic diagram of this system

c) Write the Boolean expression for this system

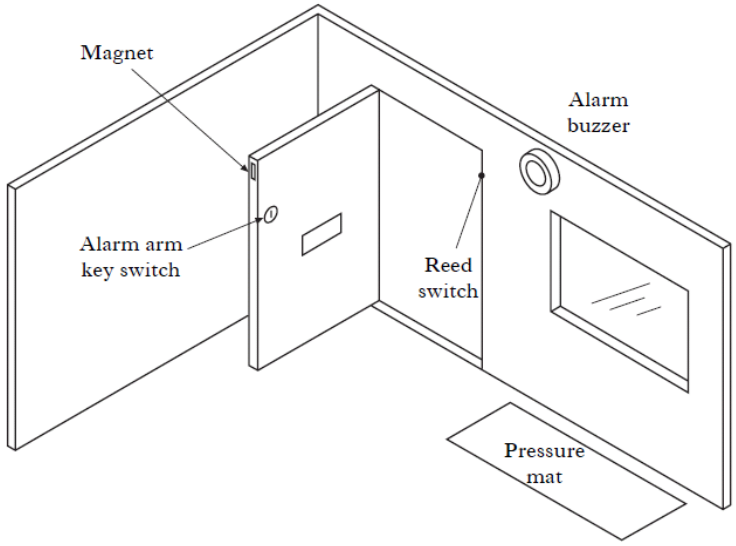
d) What is the purpose of the AND gate?

e) Why is the inverter (NOT gate) included?

16. An alarm company is developing a simple low cost security system for use in a new block of small flats.

The electronic control sub-system is based on a combinational logic circuit which operates the alarm buzzer via a transistor.

The buzzer should sound when the flat owner closes the "arm switch" **and** either the front door is forced open **or** the pressure mat below the window is stood on.



The alarm "arm switch" is incorporated into the front door key lock. Locking the door will set the alarm. A reed switch and magnet are mounted on the door and frame and this gives a low (0) signal when the front door is opened. The pressure mat gives a high (1) output when stood on.

A) Design an electronic logic control sub-system to process the signal from the inputs in order to operate the alarm buzzer as described above.

B) Use Yenka to simulate your circuit design.

Attach a print out below. If your design had to be modified in any way, explain the reason(s) why. Evaluate the system's performance against the specification.

C) Draw the truth table and work out the Boolean expression